

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Even Semester)

Innovative Practices Description

Degree, Semester & Branch: IV Semester B.E. ECE A

Course Code & Title: EC8452 Electronic Circuits II

Name of the Faculty member: Mr.A.Azhagu Jaisudhan Pazhani

Name of the Topic: Oscillators

Name of the Innovative Practice: Theoretical concepts linked with Laboratory experiments

UNIT II OSCILLATORS

Description:

Oscillators are circuits that produce an output waveform without an external signal source. The key to oscillator operation is positive feedback. A positive feedback network produces a feedback voltage V_f that is *in phase* with the input signal V_{in} as shown in Figure. The amplifier shown in the figure produces a 180° voltage phase shift, and the feedback network introduces another 180° voltage shift. This results in a combined 360° voltage phase shift, which is the same as a 0° shift. Therefore, feedback voltage is in phase with input voltage (Positive feedback can also be achieved by using an amplifier and a feedback network that *both* generate a 0° phase shift.)

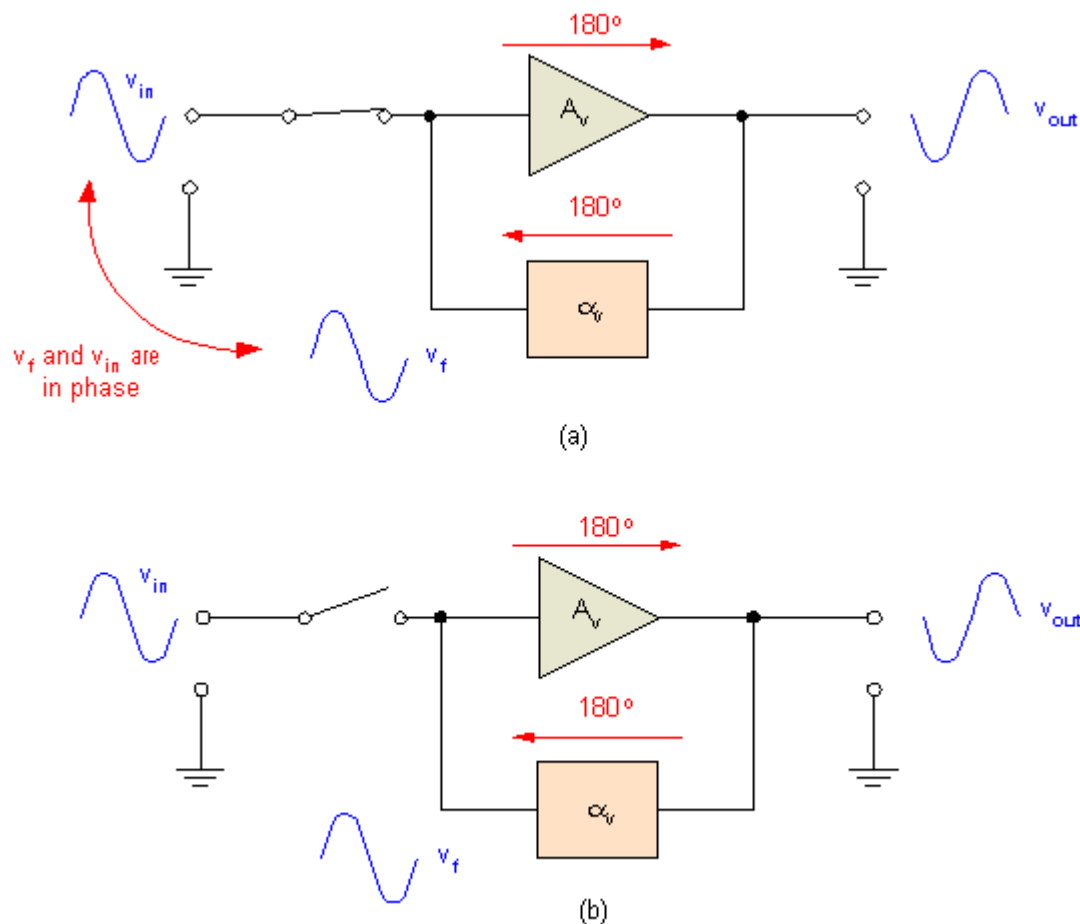


Figure: Oscillators Diagram (Regenerative Feedback)

Condition:

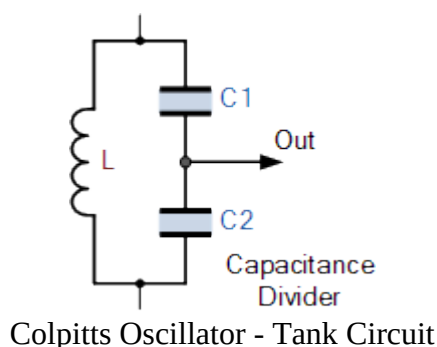
1. The circuit must have regenerative feedback; that is, feedback that results in a combined 360° (or 0°) voltage phase shift around the circuit loop.
2. The circuit must receive some trigger signal to start the oscillations.

There is one other requirement for oscillator operation. The circuit must fulfill a condition referred to as the Barkhausen criterion.

1. The loop gain is equal to unity in absolute magnitude, that is, $|A\beta| = 1$
2. The phase shift around the loop is zero or an integer multiple of 2π :

Colpitts Oscillator:

In many ways, the Colpitts oscillator is the exact opposite of the Hartley Oscillator we looked at in the previous tutorial. Just like the Hartley oscillator, the tuned tank circuit consists of an LC resonance sub-circuit connected between the collector and the base of a single stage transistor amplifier producing a sinusoidal output waveform. The basic configuration of the Colpitts Oscillator resembles that of the Hartley Oscillator but the difference this time is that the centre tapping of the tank sub-circuit is now made at the junction of a “capacitive voltage divider” network instead of a tapped autotransformer type inductor as in the Hartley oscillator.



The Colpitts oscillator uses a capacitive voltage divider network as its feedback source. The two capacitors, C1 and C2 are placed across a single common inductor, L as shown. Then C1, C2 and L form the tuned tank circuit with the condition for oscillations being: $X_{C1} + X_{C2} = X_L$, the same as for the Hartley oscillator circuit.

The advantage of this type of capacitive circuit configuration is that with less self and mutual inductance within the tank circuit, frequency stability of the oscillator is improved along with a more simple design.

In Circuits and Simulation laboratory, students are exposed to design oscillator circuit for given frequency. Therefore, I have explained theoretical concepts of oscillator which linked with laboratory experiments and also explained that how to calculate the theoretical and practical value of frequency of generated signal using oscillator circuit in order to ensure the good design of circuit.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	3	1	2	-	-	1	-	1	-	2

CO2: The student will be able to design different types of Oscillator at different frequency

References:

1. <http://www.sitttrkerala.ac.in/misc/LabManual/ECL.pdf>
2. Jacob Millman, “Microelectronics”, McGraw Hill, 2nd Edition, Reprinted, 2009.